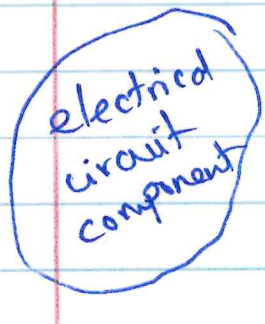
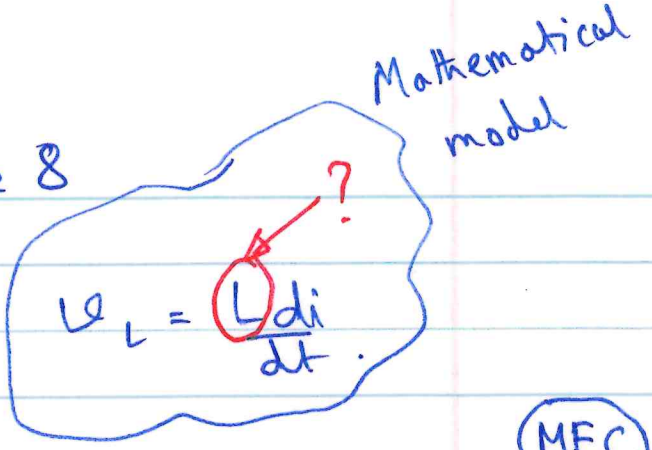
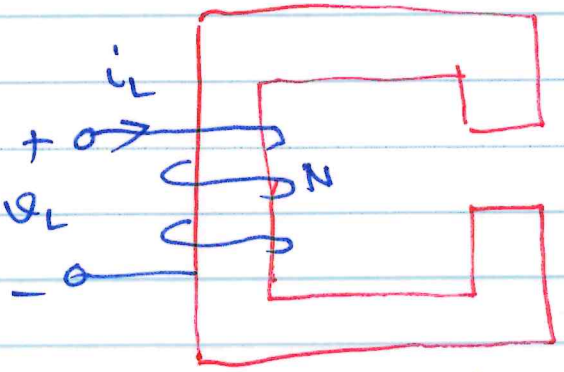
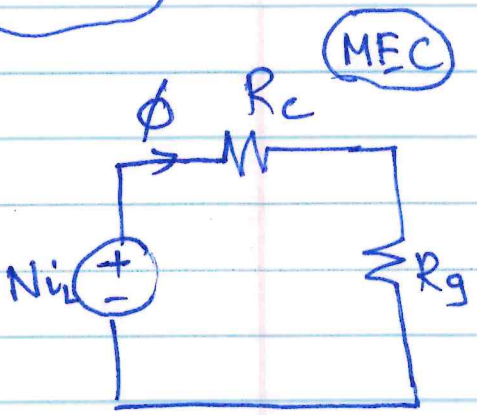
Lecture 8





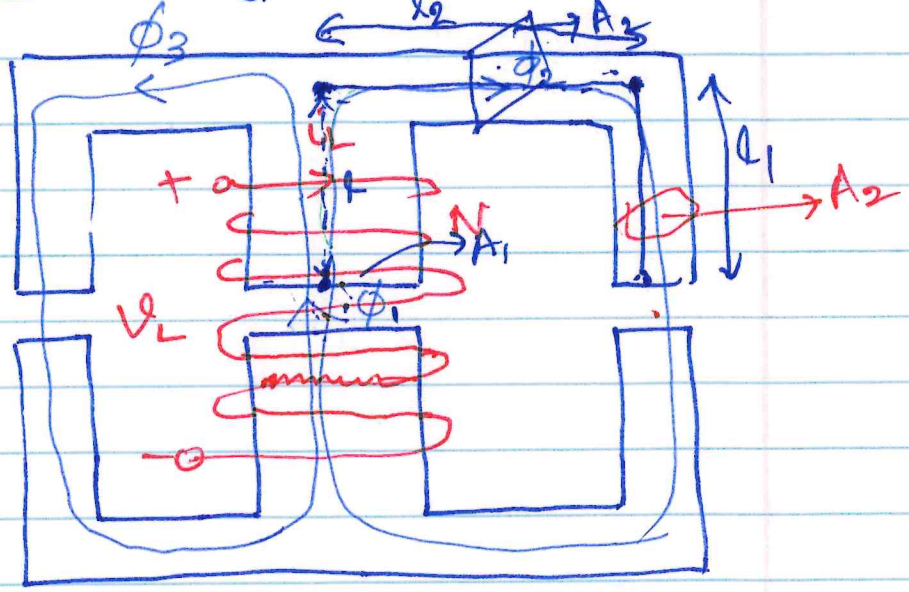


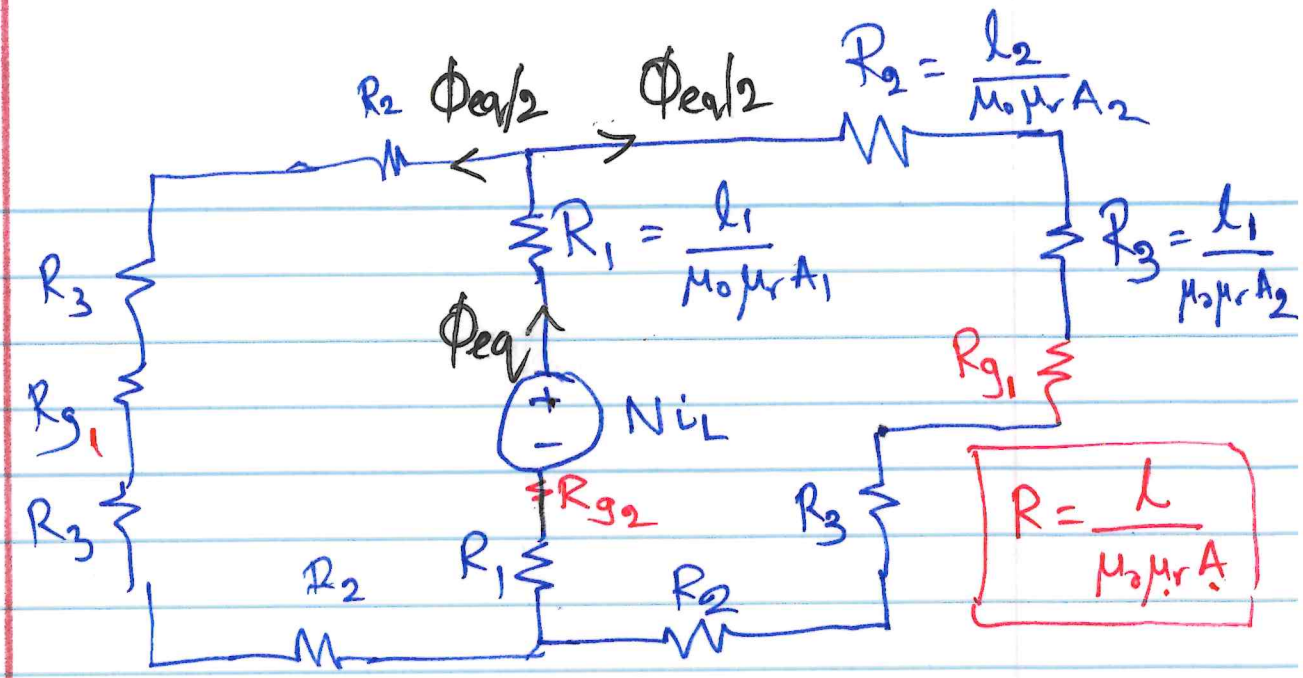
Physical Model



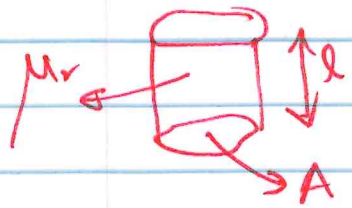
$$L = \frac{N^2}{R_{eq}}$$

Problem: Physical drawing of an inductor can ak estimate the inductance.

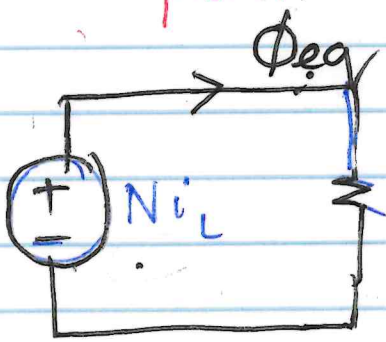




$$R_{g_1} = \frac{g}{\mu_0 A_2}$$



$$R_{g_2} = \frac{g}{\mu_0 A_1}$$

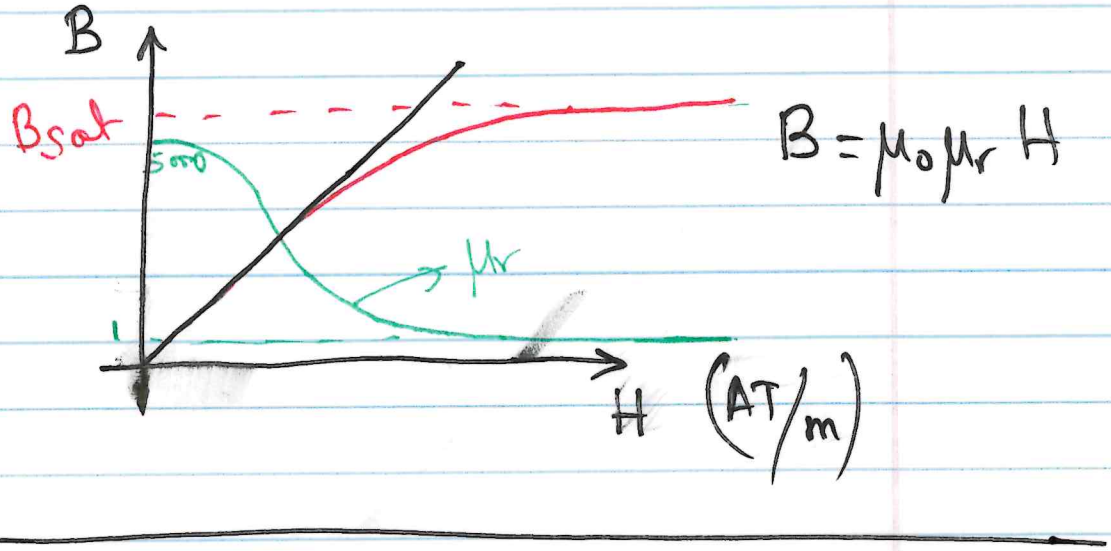


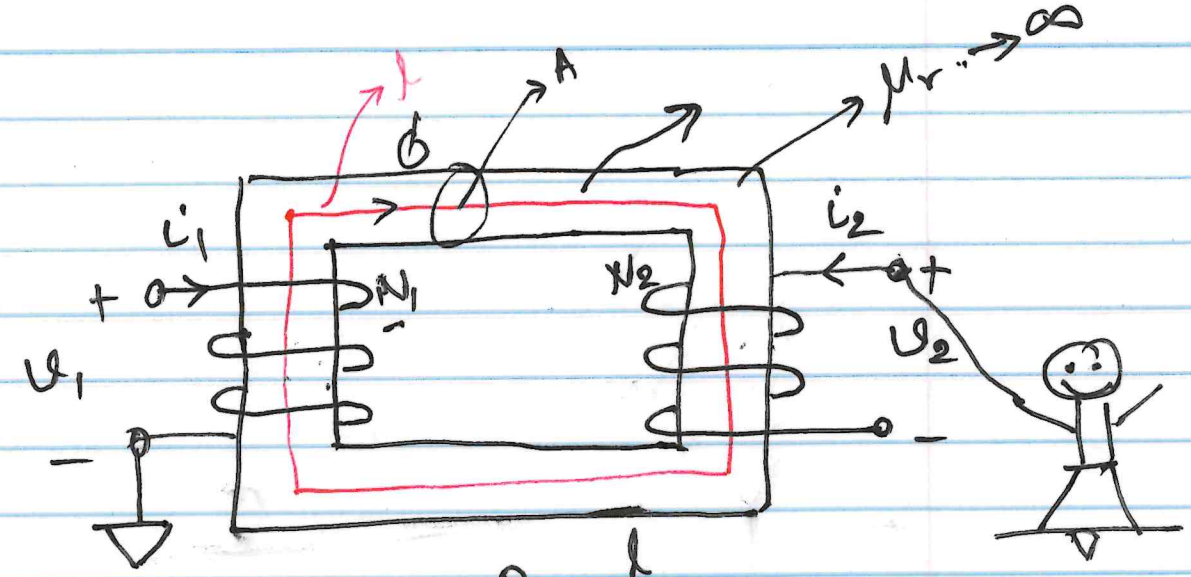
$L = ?$

$$= \frac{N^2}{R_{eq}}$$

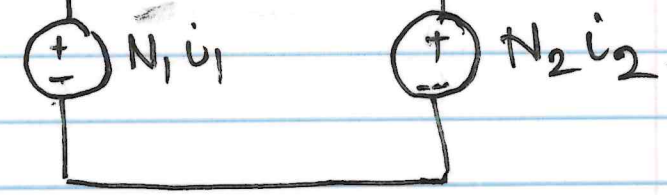
$$R_{eq} = 2R_1 + R_{g_2} + \frac{(2R_2 + 2R_3 + R_{g_1})}{2}$$

(T) Wb/m^2





$$R = \frac{l}{\mu_0 \mu_r A}$$



$$\mathcal{R} \Longrightarrow 0$$

$$\boxed{N_1 i_1 = N_2 i_2}$$

$$v_1 = N_1 \frac{d\phi}{dt} = \frac{d\lambda_1}{dt} \qquad v_2 = N_2 \frac{d\phi}{dt} = \frac{d\lambda_2}{dt}$$

$$\boxed{\frac{v_1}{N_1} = \frac{v_2}{N_2}}$$

$$p_1 = v_1 i_1 \qquad p_2 = v_2 i_2$$

$$= \left(v_2 \frac{N_1}{N_2} i_1 \right)$$

$$= v_2 \frac{N_1}{N_2} \cdot \frac{N_2}{N_1} \cdot i_2$$

$$\boxed{p_1 = p_2}$$

$$N_1 \neq N_2$$

$$v_1 = 10 \sin \omega_0 t$$

$$N_2 = 1000$$

$$N_1 = 1$$

$$v_2 = 10000 \cos \omega_0 t$$

